Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Summers, David P Ph.D.		
Project Title:	A New Ultrasensitive Technique for the Detection of Organisms and their Biomarkers		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035-1000	**Congressional District:**	22
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
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Flight Program:			
Flight Assignment:			
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COI Name (Institution):	Jahnke, Linda (NASA Ames Resarch Center)		
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This work seeks to apply a new technique, Multiphoton Detection (MPD), to the detection of biomarkers at the lowest levels. MPD can detect selected radiolabels at below background levels. By attaching radioiodine to biomarkers, in this case proteins, we can use the sensitivity of MPD to detect those biosignatures at very low levels.
The objective of this task is to apply MPD, using radiolabeling, to demonstrate the detection of proteins at very low levels. The task focuses first on developing procedures to optimize the level of detection using a variety of test proteins. This includes proteins from organisms in each of the domains (Archaea, Bacteria, and Eucarya). Work will move from the general detection of proteins to the detection of specific proteins formed in response to environmental changes. We will apply this to the detection of proteins from both live and degraded organisms from each of the domains. Finally, MPD will be applied to proteins and organisms and a variety of surface materials (including ceramics, minerals, and metals) that represent a variety of samples such as from spacecraft surfaces to planetary return samples. Limits of MPD

	protein sensitivity will be quantified.
Task Description:	This technique has a large number of potential applications to problems that span NASA's interests from the physiology of humans in space, to planetary protection, to Mars return sample handling protocols, to the search for life. For example, the detection of different Myosin isoforms occurs in the response of human muscle to weightlessness. Other proteins, undetectable by current methods, may provide even more sensitive indications of responses to weightlessness. The more sensitive such detection, the easier and sooner changes can be detected. The detection of microbial contamination is important to such issues as Planetary Protection, for both forward and back contamination. Another important aspect is the protection of spacecraft crews from exposure to undesired, or unknown, organisms or testing whether sampling handling by crews will contaminate samples. Similarly, such a technique might be used in experiments to learn more about how organisms might survive exposure to space. More generally, the search for life, and the biosignatures that it leaves behind, is a central mandate of Astrobiology and the study of Martian return samples. Protein molecules are present in all organism and their detection is central to NASA's missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)